Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / Bending
Case	85ecb8b8-6e95-45d9-a022-51dca0494ba8
Date	2026-11-01 18:02:46
Status	Reviewed by expert (expert)

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	92%	
dent	major	88%	
crack	critical	76%	
deformation	major	91%	

- L2: -0.11 mm
- H1: -0.05 mm

2. Root cause

Cause: Tool Wear and Die Damage

Reason: Reason: The detected minor scratch (32.0 mm2), major dent (64.0 mm2), and dimensional deviations on edge L2 (-0.114 mm) and hole H1 (-0.054 mm) match evidence [seed-003], which attributes scratches, dents, and dimensional variations to worn tooling and die damage.

Effect: Surface Scratches, Dents, Dimensional Variation

Confidence: 68%

Supporting evidence
- **seed-003**: Tool Wear and Die Damage cause scratches, dents and dimensional variations. Preventive actions: regular tooling inspection, preventive maintenance schedule, die regrinding.

3. Solution

Corrective actions (fix this occurrence)

1. Recalibrate press brake
2. Verify material lot

Preventive actions (stop it recurring)

1. Schedule monthly calibration
2. Incoming inspection

Precautions

- Check die condition before starting production run

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Material Inconsistency and Hardness Variation	65%	Reason: The presence of a critical crack (56.0 mm2) and major deformation (400.0 mm2) on the sheet metal bracket points to material hardness variation and inconsistency, which cause unpredictable deformation and cracking according to [seed-002].

3	Improper Bending Force or Over Bending	62%	Reason: The detected major deformation (400.0 mm2) and negative dimensional deviations on features L2 (measured 79.886 mm vs nominal 80.0 mm) and H1 (measured 11.946 mm vs nominal 12.0 mm) correspond directly to improper bending force or over bending as documented in [seed-001].

5. Traceability

AI model: gemini:gemini-3.5-flash. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.